$$Z = \frac{X-\mu}{\sigma} = \frac{145-150}{15} = -0.33$$

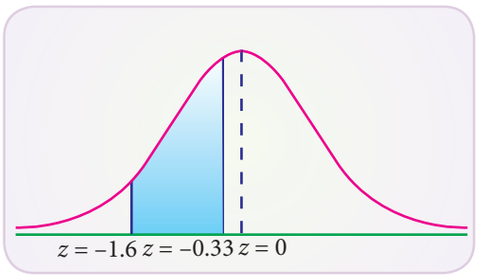


Fig 7.10

Area between $Z = 0$ and $Z = -1.67$ is 0.4525

Area between $Z = 0$ and $Z = -0.33$ is 0.1293

$P(-1.667 \le Z \le -0.33) = 0.4525 - 0.1293$

$= 0.3232$

Therefore the number of branches having sales between ₹ 1,25,000 and ₹ 1,45,000 is $550 \times 0.3232 = 178$

(ii) When $X = 140$ thousand

$$Z = \frac{X-\mu}{\sigma} = \frac{140-150}{15} = -0.67$$

When $X = 160$ thousand

$$Z = \frac{X-\mu}{\sigma} = \frac{160-150}{15} = 0.67$$

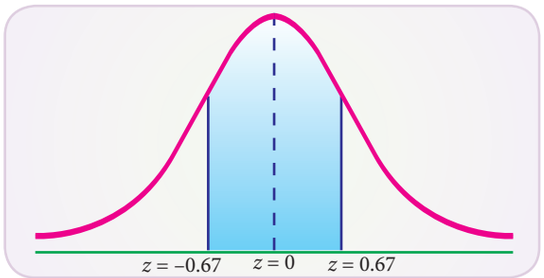


Fig 7.11

$P(-0.67 < Z < 0.67)$

$= P(-0.67 < Z < 0) + P(0 < Z < 0.67)$

$= P(0 < Z < 0.67) + P(0 < Z < 0.67)$

$= 2\,P(0 < Z < < 0.67)$

$= 2 \times 0.2486$

$= 0.4972$

Therefore, the number of branches having sales between ₹ 1,40,000 and ₹ 1,60,000

$= 550 \times 0.4972 = 273$

Example 7.24

Assume the mean height of children to be 69.25 cm with a variance of 10.8 cm. How many children in a school of 1,200 would you expect to be over 74 cm tall?

Solution

Let the distribution of heights be normally distributed with mean 68.22 and standard deviation = 3.286

$$Z = \frac{X-\mu}{\sigma} = \frac{X-69.25}{3.286}$$

When $X = 74$

$$Z = \frac{X-\mu}{\sigma} = \frac{74-69.25}{3.286} = 1.4455$$

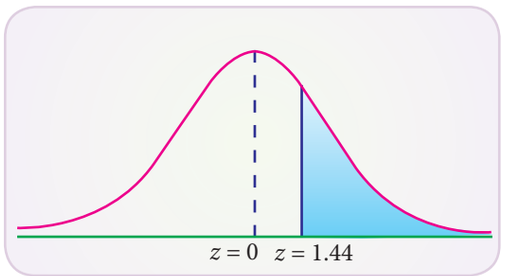


Fig 7.12

Now $P(Z > 74) = P(Z > 1.44)$

$= 0.5 - 0.4251$

$= 0.0749$

Expected number of children to be over 74 cm out of 1200 children

$= 1200 \times 0.0749 \approx 90$ children

XII Std - Business Maths & Stat EM Chapter 7.indd 164 2/29/2020 11:34:34 AM